

BLA-03 · BUILDING AI APPS

Connecting, Shipping and Running Your AI App

Letting parts talk without waiting · Shipping changes safely · Tracking experiments · The front end · Where it runs

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PART

03

Letting parts talk without waiting

Service	Think of it as	Use it when
SQS	A to-do list	Work can wait its turn
SNS	A loudspeaker	Many people need the same message
EventBridge	A post room	Different events go to different places
AppFlow	A courier	Moving data from apps like Salesforce



If one part doesn't have to wait for another, a slow part can't hold up the rest.

A real example

 Monday morning: 10,000 contracts are uploaded at once.



 **Without the queue,** 10,000 requests hit the model at once — and most get rejected.

Shipping changes safely

CI/CD: a conveyor belt from your code to production.



CodePipeline runs the whole belt, start to finish — nobody copies files around by hand.

💡 **A prompt or model setting is a change too.** It can break things just as easily as code — so it deserves the same tests.

Tracking experiments

MLflow keeps a record of every experiment you try.

Try	Prompt	Model	Temperature	Score
1	v1	Large	0.7	72%
2	v2	Large	0.2	81%
3	v2	Small	0.2	80%



Try 3 is nearly as good as Try 2 — and much cheaper.

You only know that because you wrote it down.

The front end

The part people actually see.

Service	What it's for
Amplify	Build and host the web or mobile app people use
AppSync	Push updates to the app in real time



Example: a chat screen where the answer appears word by word — AppSync pushes each piece as the model writes it.

Where it all runs

Service	What it is	Use it when
AWS Regions	The normal cloud	Almost always
Outposts	AWS hardware in your building	Data can't leave your site
Wavelength	AWS at the edge of 5G networks	Every millisecond counts

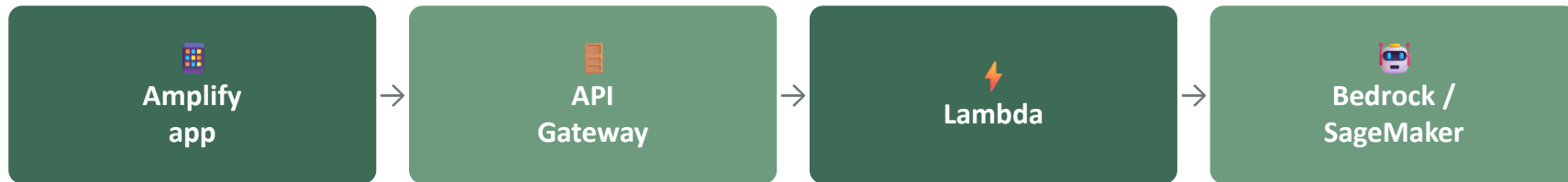


A hospital that must keep patient data on site → Outposts



A live translation app on a phone → Wavelength

Putting it all together



Around the request path	Its job
SQS · SNS	Queue big jobs, tell people when they're done
AppConfig · Step Functions	Pick the model, chain the prompts
CodePipeline · MLflow	Ship changes safely, record what worked

My three takeaways

- 1 The model is one box in a much bigger diagram
- 2 Queues protect the model from sudden floods
- 3 A prompt change is a code change — test it, roll it out carefully